WORKSHEET - 5

Course : Aptitude **Topic:PROIT AND LOSS**

EXERCISE

1. If C.P. is ` 2516 and S.P. is ` 2272, find the percentage loss. (R.R.B., 2006)

Sol. C.P. = ` 2516, S.P. = ` 2272, Loss = ` (2516 – 2272) = ` 244.

Loss% = (244/2516 × 100)% = 9.69%

2. Find S.P., when (i) C.P. = ` 56.25, Gain = 20% (ii) C.P. = ` 80.40, Loss = 15%

Sol. (i) S.P. = 120% of ` 56.25 = (120/100 × 56.25) = 67.50.

(ii) S.P. = 85% of ` 80.40 = (85/100 × 80.40)= 68.34.

3. A gold bracelet is sold for ` 14500 at a loss of 20%. What is the cost price of the gold bracelet? (Bank Recruitment, 2008)

Sol. C.P. = (100/80 × 14500) = 18125.

4. A television manufacturer earns 20% profit by selling each T.V. set for ` 14400. If the production cost is increased by 15%, what should be the new selling price of a set so as to gain 15%? (S.S.C., 2006)

Sol. S.P. = 14400, Profit = 20%.

C.P. = (100/120× 14400) = 12000.

New C.P. =(115/100 ×12000)=13800 ; Gain = 15%.

Required S.P. = (115/100 × 13800) = 15870.

5. A book was sold for ` 27.50 with a profit of 10%. If it were sold for ` 25.75, then what would have been the percentage of profit or loss?

Sol. S.P. = ` 27.50, Profit = 10%.

So, C.P. = (100/110 × 27.50) = ` 25.

When S.P. = ` 25.75, profit = ` (25.75 – 25) = ` 0.75.

Profit% = (0.75/25 × 100))% = 3%.

6. If the cost price is 96% of the selling price, then what is the profit percent ?

Sol. Let S.P. = ` 100. Then, C.P. = ` 96; Profit = ` 4.

Profit% = (4/96 ×100)% = 25/6 % = 4.17%

7. A manufacturer makes 800 articles at a cost of ` 1.50 per article. He fixes the selling price such that if only 600 articles are sold, he would make a profit of 30% on his outlay. However, he sold 620 articles at this price. Find his actual profit percent of the total outlay, assuming that the unsold articles are useless. (P.C.S., 2006)

Sol. C.P. of 800 articles = ` (1.50 × 800) = ` 1200.

S.P. of 600 articles = (130/100 × 1200) = ` 1560.

S.P. of 620 articles = (1560/600 × 620) = ` 1612.

Profit = ` (1612 – 1200) = ` 412.

Profit % = (412/1200×100)%= 34 1/3 %

8. The selling price of 30 items is equal to the purchase price of 25 items. What is the profit or loss percent? (Bank Recruitment, 2007)

Sol. Let C.P. of each article be ` 1.

Then, C.P. of 30 items = ` 30,

S.P. of 30 items = ` 25.

Loss = ` (30 – 25) = ` 5.

Loss% = (5/30 ×100) % = 16 2/3 %.

9. By selling 33 metres of cloth, one gains the selling price of 11 metres. Find the gain percent.

Sol. (S.P. of 33 m) – (C.P. of 33 m) = Gain = S.P. of 11 m.

S.P. of 22 m = C.P. of 33 m.

Let C.P. of each metre be ` 1. Then, C.P. of 22 m = ` 22, S.P. of 22 m = ` 33.

Gain% = (11/22 ×100)% = 50%

10. A vendor bought bananas at 6 for ` 10 and sold them at 4 for ` 6. Find his gain or loss percent.

Sol. Suppose, number of bananas bought = L.C.M. of 6 and 4 = 12.

C.P. = (10/6 × 12) = ` 20; S.P. = (6/4 × 12) = ` 18.

Loss% = $(\frac{2}{20}\times 100)\%$ =10%

11. A grocer purchased 80 kg of sugar at ` 13.50 per kg and mixed it with 120 kg sugar at ` 16 per kg. At what rate should he sell the mixture to gain 16%?

Sol. C.P. of 200 kg of mixture = ` (80 × 13.50 + 120 × 16) = ` 3000.

S.P. = 116% of ` 3000 = (116/100 × 3000) = 3480.

Rate of S.P. of the mixture = (3480/200) per kg = ` 17.40 per kg.

12. A dishonest dealer professes to sell his goods at cost price but uses a weight of 960 gms for a kg. weight. Find his gain percent.

Sol. Gain % = $(\frac{40}{960}\times 100)$ % = 4 1/6 %

13. A shopkeeper advertises for selling cloth at 4% loss. However, by using a false metre scale he actually gains 25%. What is the actual length of the scale? (R.R.B., 2008)

Sol. Let C.P. of 1 m of cloth be ` 1.

Professed S.P. of 1 m cloth = (96 /100) = ` 0.96. Gain = 25%.

Actual C.P. of the cloth sold for ` 0.96 = `(100× 0.96/125) = ` 0.768.

Actual length of the scale = Length bought for ` 0.768 = 76.8 cm.

14. A dishonest dealer sells the goods at 1 6 %4 loss on cost price but uses 1 12 %2 less weight. What is his percentage profit or loss?

Sol. Let the C.P. of 1 kg goods be ` 1.

Then, S.P. of [(100 – 12 ½)% of 1 kg] i.e., 875 g goods = ` 0.9375.

S.P. of 1 kg goods = (0.9375× 1000/875) = ` 1 1 /14

Profit % = (1×100 / 14)% = 7 1/7 %

15. If the manufacturer gains 10%, the wholesale dealer 15% and the retailer 25%, then find the cost of production of a table, the retail price of which is ` 1265 ?

Sol. Let the cost of production of the table be ` x.

Then, 125% of 115% of 110% of x = 1265

⇒253x/160 = 1265

x = 800

16. An article is sold at a certain price. By selling it at 2 3 of that price one loses 10%. Find the gain percent at original price.

Sol. Let the original S.P. be ` x.

Then, New S.P. = ` 2 /3 x ,Loss = 10%.

So,C.P.= (100/90 × 2x/3) = 20x/27

Now, C.P. = ` 20x/27 ,S.P. = ` x. Gain = (x – 20x/27) = 7x/27

Gain% = (7x/27 × 27/20x ×100)% = 35%.

17. A tradesman sold an article at a loss of 20%. If the selling price had been increased by ` 100, there would have been a gain of 5%. What was the cost price of the article?

Sol. Let C.P. be ` x. Then, (105% of x) – (80% of x) = 100 or 25% of x = 100

x/4 = 100 or x = 400.

So, C.P. = ` 400.

18. A man sells an article at a profit of 25%. If he had bought it at 20% less and sold it for ` 10.50 less, he would have gained 30%. Find the cost price of the article.

Sol. Let the C.P. be ` x.

1st S.P. = 125% of x = 125x/100 = 5x/4, 2nd C.P. = 80% of x = 80x/100 = 4x/5

2nd S.P. = 130%of 4x/5 = (130/100 × 4x/5) = 26x/25

21x/100 = 10.50

X = 50

Hence C.P. = 50.

19. The price of a jewel, passing through three hands, rises on the whole by 65%. If the first and the second sellers earned 20% and 25% profit respectively, find the percentage profit earned by the third seller.

Sol. Let the original price of the jewel be ` P and let the profit earned by the third seller be x%.

Then, (100 + x)% of 125% of 120% of P = 165% of P $\Rightarrow$X = 10%

20. Find the single discount equivalent to a series discount of 10%, 20% and 30%.

(S.S.C.)

Sol. Let marked price be ` 100.

Then, Net S.P. = (100 – 30)% of (100 – 20)% of (100 – 10)% of ` 100

= 70% of 80% of 90% of ` 100

= ` 50.40.

Required discount = (100 – 50.40)% = 49.6%.

21. After getting two successive discounts, a shirt with a list price of ` 150 is available at ` 105. If the second discount is 12.5%, find the first discount.

Sol. Let the first discount be x%.

Then, 87.5% of (100 – x)% of 150 = 105

⇒ x = (100 – 80) = 20.

First discount = 20%.

22. An uneducated retailer marks all his goods at 50% above the cost price and thinking that he will still make 25% profit, offers a discount of 25% on the marked price. What is his actual profit on the sales?

Sol. Let C.P. = ` 100. Then, marked price = ` 150.

S.P. = 75% of ` 150 = ` 112.50.

Gain% = 12.50%.

23. A retailer buys 40 pens at the marked price of 36 pens from a wholesaler. If he sells these pens giving a discount of 1%, what is the profit percent?

Sol. Let the marked price of each pen be ` 1.

Then, C.P. of 40 pens = ` 36. S.P. of 40 pens = 99% of ` 40 = ` 39.60.

Profit % = (3.60 × 100/36)%= 10%.

24. The market price of a radio is ₹. 480. The shopkeeper allows a discount of 10% and gains 8%. If no discount is allowed, find is Gain percentage

Sol. Market price is 480, Discount is 10%= $480 \times \frac{10}{100} = 48$

Then Selling price = M.P – Discount = 480 – 48 = 432

Cost price = $432 \times \frac{100}{108} = 400$

If no discount then S.P = 480, Profit% = $\frac{80}{400} \times 100 = 20\%$

25. A shopkeeper sold a watch for ₹. 144 and got a percentage of profit equal to the cost price. Find the cost price of the watch.

Sol. Let the cost price of the watch $= ₹.X$

$$\text{Then } X\left(\frac{100+X}{100}\right) = ₹.144$$

$$(X + 180)(X - 80) = 0$$

$$X = -180 \text{ or } 80.$$

Only positive value should be the cost price.

$$\text{So CP} = ₹ 80$$

PRACTICE QUESTIONS

1. A dealer buys an article marked at ₹. 25000 with 20% and 5% off. He spends ₹. 1000 on its repairs and sells it for ₹. 25000. What is his gain or loss percent?

(S.S.C., 2007)

(a) Loss of 25% b) Gain of 25% c) Loss of 10% d) Gain of 10%

2. If an article with marked price of ₹. 400 is sold at successive discounts of 10%, 25% and 15%, what is the approximate price the customer has to pay?

(Campus Recruitment, 2009)

a) ₹. 230 b) ₹. 270 c) ₹. 300 d) ₹. 360

3. A trader marked the price of a product in such a way that it is 20% more than the cost price. If he allows 10% discount on the marked price to the customer then his gain is

(L.I.C.A.D.O., 2008; S.S.C., 2005, 06; C.P.O., 2007; Hotel Mgmt, 2007)

a) 8% b) 10% c) 15% d) 20%

4. A trader marked the price of his commodity so as to include a profit of 25%. He allowed discount of 16% on the market price. His actual profit was **(S.S.C., 2004)**

(a) 5% b) 9% c) 16% d) 25%

5. By selling an umbrella for ₹. 300, a shopkeeper gains 20%. During a clearance sale, the shopkeeper allows a discount of 10% on the market price. His gain percent during the sale is **[TCS]**

a) 7 b) 7.5 c) 8 d) 9

6. The cost price of an article is 64% of the marked price. Calculate the gain percent after allowing a discount of 12%.

a) 37.5% b) 48% c) 50.5% d) 52%

7. The marked price is 10% higher than the cost price. A discount of 10% is given on the market price. In this kind of sale, the seller

[SSC—CHSL (10 + 2) Exam, 2015]

a) gains 2% b) bears no loss, no gain c) gains 1% d) loses 1%

8. The profit earned by selling a article at ₹. 5520 is equal to the loss incurred on selling the same article at ₹. 4080. What will be percent profit, if the article is sold at ₹. 6000?

[United India Insurance (UIICL) Assistant (Online) Exam, 2015]

a) 20 b) 12 c) 25 d) 15

9. Sanjay made a profit of 8% by selling a shirt after offering a discount of 12%. If the marked price of the shirt is ₹. 1080, find its cost price.

[SBI Jr. Associates (Pre.) Exam, 2016]

a) 890 b) 780 c) 880 d) 900

10. Supriya sold a washing machine for ₹. 8500. She incurred a loss of 15% in this transaction. At what price had she bought the washing machine?

[ESIC—UDC Exam, 2016]

a) ₹. 10000 b) ₹. 1200 c) ₹. 11000 d) ₹. 10500

11. The price of a cycle is marked at ₹. 1150. A shopkeeper earns a profit of 15% after allowing a discount of 15% on the market price. Find the cost price of the cycle.

[ESIC—UDC Exam, 2016]

(a) ₹. 900 b) ₹. 1000 c) ₹. 850 d) ₹. 950

12. A man bought apples at the rate of 30 for ₹.100/-. How many should he sell for ₹.50/- as to gain 50%?

a) 5 b) 10 c) 15 d) 20

13. A purchased a car for ₹.50,000/- and sold to B at a gain of 4%. B sold it to C at a gain of 6%. What C pays for it? **[AMAZON]**

a) ₹.55,000 b) ₹.52,000 c) ₹.52,120 d) ₹.55,120

14. An article costing ₹.84/- was sold at a profit of 50%. The second purchaser sold it once again at a loss of 25%. At what price did he sell? **[WIPRO]**

a) ₹.90 b) ₹.92 c) ₹.98.50 d) ₹.94.50

15. A retailer increases the price of an article first by 50% and then reduces price by 50%. What is percent $\frac{\text{gain}}{\text{loss}}$?

a) 25% gain b) 25% loss c) 20% gain d) 20% loss

16. A man sells two radio sets for ₹.250 each. At one he gains 10% and on the other he loses 10%. What is his total gain or loss? [**Amazon]**

a) ₹.5.05 loss b) ₹.5.05 gain c) Can't say d) None of these

17. A man sells 2 horses at the rate of ₹.1,500 each. At one he gains 25% and on the other he loses 25%. What is his total gain or loss percentage?
[TCS]

a) ₹.200 gain b) ₹.200 loss c) Can't say d) None of these

18. A man sells two tables at same price, on one he makes a profit of 50% and on the other he loses 50%; on the whole whether he gets profit or loss; how much %?

a) 25% gain b) 25% loss c) 50% loss d) No loss no gain

19. An electric iron was sold at a profit of 15%, had it been sold for ₹.600/- the profit would have been 20%. What is the former selling price of electric iron? **[WIPRO]**

a) ₹.500 b) ₹.575 c) ₹.600 d) ₹.675

20. A man sold his watch at a loss of 5%. Had he sold it for ₹.56.25 more he would have gained 10%. What is the cost price of the watch?
[ACCENTURE]

a) ₹.300 b) ₹.360 c) ₹.375 d) ₹.400

21. An article when sold at a gain of 5% yields ₹.15 more than when sold at a loss of 5%. What was its cost price?

a) ₹.100 b) ₹.120 c) ₹.125 d) ₹.150

22. Dayanand purchased 150 kg of wheat at the rate of ₹.700 per quintal. He sold 50 kg and made 10% profit. At what rate per kg should he sell the remaining to get profit of 20% on the total deal? **[WIPRO]**

a) ₹.6 b) ₹.7.50 c) ₹.8.75 d) ₹.9

23. A trader purchased 250 kg of rice at the rate of ₹.500 per quintal. He sold 100 kg and was lost 5%. At what rate per kg should he sell the remaining to get profit of 10% on the total deal?
[INFOSYS]

a) ₹.5 b) ₹.6 c) ₹.8 d) ₹.9.25

24. A man has goods worth ₹.300; he sells one third of them so as to lose 10%. By how much percent should he raise the selling price in order to gain 10% on the whole?

a) 10% b) 20% c) 30% d) 40%

25. The price of sugar being raised by 20%. By how much percent must reduce his consumption of that article so as not to increase his expenditure? **[TCS]**

a) $33\frac{1}{3}$ % b) $16\frac{2}{3}$% c) 20% d) 40%

KEY

	1	2	3	4	5	6	7	8	9	10
1-10	B	A	A	A	C	A	D	C	C	A
11-20	C	B	D	D	B	A	B	B	B	C
21-25	D	C	B	B	B					